Ecology and Ethology

34th International Biology Olympiad

3-10 July 2023, United Arab Emirates University

Practical Exam

Total points: 103

Duration: 90 minutes

General Instructions:

You have 90 minutes to complete **FOUR tasks in this practical exam.**

You can do the tasks in any order. Note that the formulae for standard deviation and standard error have been provided in Q.1.3.

Task 1: Population estimation (17 points)

Task 2: Effect of guano on invertebrates (27 points)

Task 3: Thermoregulation in cormorants (32 points)

Task 4: Sexual communication in lizards (28 points)

Write your answers in the answer sheet. Only answers given in the answer sheet will be evaluated.

Make sure that you have received all the materials and equipment listed. If any of these items are missing, please raise your card immediately.

Do not use any other applications or browsers during this exam.

If you face any technical issues with your computer, raise your card.

The speed of the videos in Tasks 3 and 4 can be changed by right clicking on the video in Media Player and selecting the appropriate speed of your choice.

During the videos, make sure to keep your cursor on the Media Player task bar.

Use the following cards to ask for water/washroom/help.

Drinking Water	Washroom	Other Queries
		?

Stop answering as soon as you hear the whistle at the end of the exam.

No paper, materials or equipment should be taken out of the laboratory.

Good luck!

You are provided with

1. A bag of beads
2. A marker pen
3. A counter

Before you start, check the items.

Task 1. Mark-Release-Recapture Method to Estimate Population Size

Mark-release-recapture methods are used to estimate the population size of mobile animals. A known number of individuals of a population are captured and marked, with a ring or an ear tag, which has been tested to not cause any mortality, and then released back into the population. When individuals of the same population are captured again, after giving the individuals sufficient time to mix with the rest of the population, the researchers expect to recapture a certain proportion of the previously marked individuals. The marked individuals represent a proportion of the total population that can be used to estimate the total population of individuals.

Q.1.1 Given

- "N" is the estimated total population size,
- "M" is the number of individuals captured and marked,
- "C" is the number of individuals captured successively with or without a mark,
- "R" is the number of marked individuals recaptured.

Q.1.1	Write down the formula by which "N" can be estimated.	2.0pt

Q.1.2 If no marked individuals are recaptured through the above process, we conclude that:

Q.1.2 **True or False?** 3.0pt

Statement	**True**	**False**
A. The population is infinitely large		
B. The marked individuals have all died		
C. It is necessary to mark more individuals and make fresh attempts to estimate the size of the population		

Q.1.3 In your first task, you have a bag, containing a known number of beads, to simulate a population of animals.

1. Take out 20 beads randomly, mark them with the marker provided and then return the marked beads to the bag.
2. Shake the bag well to ensure that the beads are mixed properly.
3. Take out another 20 beads randomly and note how many of them are already marked from your first capture.
4. Mark the unmarked beads from this second capture and then return all the newly marked and previously marked beads back to the bag.
5. Conduct this process for a total of five times. In each event, mark the **unmarked** beads that you capture and return all beads, so that the total number of marked beads increases with each repeat round.
6. Fill the table. Estimate the total population size, mean population estimate and the standard error of the estimate to one decimal place.

Standard deviation,

$$\boxed{\mathrm{SD} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}}$$

Standard error,

$$\boxed{\mathrm{SE} = \frac{\mathrm{SD}}{\sqrt{n}}}$$

x_i	the value of the i-th point in the data set
$\bar{x}$	The mean of the data set
n	The number of data points in the data

Q.1.3.1 4.0pt

Marking event	Numbers of individuals captured with or without a mark (C)	Number of marked individuals recaptured (R)	Number of individuals newly marked	Total number of individuals captured and marked (M)	Estimated total population size (N)
1	20	-	20	20	-
2	20				
3	20				
4	20				
5	20				
Total	100	X	X	X	X

Q.1.3.2 1.0pt

Mean population estimate	

Q.1.3.3 1.0pt

Standard error of the estimate	

Mark with a cross (X) the correct answer

Q.1.3.4 1.0pt

	The final estimate	The mean of the estimates
Which would finally be a better estimate of the total population size?		

In a real population of marked animals, such as the Arabian red fox *Vulpes vulpes arabica*, shown above, various factors can introduce errors in the estimated population size.

Considering the process described above, indicate which of the following factors would introduce an error in the population estimate.

Q.1.3.5 True or False? 5.0pt

Factor	**True**	**False**
A. The same individual fox is trapped more frequently than others		
B. The collar on an individual fox drops off		
C. The collared foxes do **not** emigrate during the experiments		
D. A very small proportion of the fox population is marked at the end of all mark-recapture events		
E. The capturing events are carried out in very rapid succession		

Task 2. Response of Terrestrial Invertebrates to Guano Enrichment from Nesting Cormorants

The Socotra cormorant *Phalacrocorax nigrogularis*, is a threatened aquatic bird, endemic to the Arabian Gulf and found off the southeast coast of the Arabian peninsula. They usually nest in colonies of tens-to-thousands of nesting pairs. In the breeding season, between August and December, they build cup-shaped nests on the ground, lay eggs, incubate them and tend to their hatchlings, including protecting them from predators, such as the nocturnal Arabian red fox.

Large colonies of breeding Socotra cormorants deposit thousands of tons of faeces (guano), which alter soil chemistry, greatly increasing nutrients, such as nitrogen. This has variable impacts on the soil flora and fauna. Population of some species increase due to the guano enrichment, others decrease, and some remain unaffected.

You are given data showing invertebrates captured in pitfall traps that were placed in a large cormorant colony. Pitfall traps are small, cup-shaped containers, buried up to their rim in the ground, with a liquid preservative in them to prevent insects from escaping when they fall in.

The image above shows a cormorant nesting area (left) and a non-nesting area (right). Pitfall traps (blue dots) were placed in both areas for one week in the month before the nesting period. The pitfall traps were put back in the same place for another week during the month after the nesting period. The trap contents were collected at the end of the one-week periods. All the collected invertebrates were then identified and counted. These taxa included Coleoptera (beetles), Araneae (spiders), Formicidae (ants) And Argasidae (soft ticks). Their numbers have been reported in Tables 1 and 2 below.

Table 1

Nesting Area – Before Breeding Season					Nesting Area – After Breeding Season				
Pitfall Traps	Beetles	Spiders	Ants	Ticks	Pitfall Traps	Beetles	Spiders	Ants	Ticks
Trap 1	33	13	10	2	Trap 1	4	1	1	38
Trap 2	27	17	11	5	Trap 2	7	2	1	41
Trap 3	29	18	9	1	Trap 3	4	2	2	32
Trap 4	19	18	12	3	Trap 4	3	3	1	45
Trap 5	22	14	9	4	Trap 5	1	1	1	28
Trap 6	26	19	8	4	Trap 6	8	4	3	27
Trap 7	31	15	11	3	Trap 7	5	6	1	31
Trap 8	26	14	13	2	Trap 8	2	2	1	33
Trap 9	21	19	12	7	Trap 9	2	3	1	37
Trap10	28	20	11	2	Trap 10	1	3	2	41
Mean					Mean	4.1	2.7	1.4	35.3
Standard deviation					Standard deviation	2.4	1.5	0.7	6.0

Table 2

Non-Nesting Area – Before Breeding Season					Non-Nesting Area – After Breeding Season				
Pitfall Traps	Beetles	Spiders	Ants	Ticks	Pitfall Traps	Beetles	Spiders	Ants	Ticks
Trap 1	25	15	7	3	Trap 1	3	13	14	1
Trap 2	22	16	10	2	Trap 2	2	17	11	4
Trap 3	20	18	11	1	Trap 3	1	19	12	2
Trap 4	18	18	13	2	Trap 4	1	13	10	3
Trap 5	18	19	10	6	Trap 5	5	14	10	3
Trap 6	17	21	11	1	Trap 6	4	13	12	5
Trap 7	19	14	13	1	Trap 7	2	18	11	4
Trap 8	15	12	9	2	Trap 8	2	16	12	4
Trap 9	20	13	13	5	Trap 9	7	15	10	1
Trap 10	21	13	12	1	Trap10	1	13	9	2
Mean	19.5	15.9	10.9	2.4	Mean	2.8	15.1	11.1	2.9
Standard deviation	2.8	3.0	2.0	1.8	Standard deviation	2.0	2.3	1.4	1.4

Q.2.1	Calculate the mean and standard deviation of the abundance of beetles, spiders, ants, and ticks in the Nesting Area - Before Breeding Season and write them, rounded to one decimal place in the table.	8.0pt

Q.2.2

Calculate the Shannon-Wiener Diversity Index (H') for Trap 1 of pitfall traps in the Nesting Area – Before Breeding Season and in the Nesting Area – After Breeding Season, by first using the following formula:

$$H' = -\sum_{i=1}^{S} p_i \ln p_i$$

where p_i, the proportion of taxon 'i' relative to the total number of taxa, is computed, and then multiplied by ln p_i, the natural logarithm of this proportion.

Give the answer, rounded to two decimal places, in the boxes below:

Q.2.2.1	Nesting Area – Before Breeding Season	2.0pt

Q.2.2.2	Nesting Area – After Breeding Season	2.0pt

Q.2.3 True or False?

Q.2.3.1 The diversity of taxa, as measured by H', is higher in the Nesting Area – After Breeding Season. 1.0pt

True	
False	

Q.2.3.2 In general, the H' for a habitat, with only a single species present, is 0. 1.0pt

True	
False	

Q.2.3.3 H' depends on both species richness, which is the number of species in a particular habitat, and on species evenness, which is the relative abundance of each of the species in that habitat. 1.0pt

True	
False	

Q.2.3.4 The H' of a particular habitat decreases as species evenness, in that habitat, increases. 1.0pt

True	
False	

Q.2.3.5 A habitat with a relatively higher H' value, compared to another habitat, is likely to have lower interspecies competition for resources. 1.0pt

True	
False	

Q.2.4 True or False?

Q.2.4.a The observed differences in arthropod abundance before and after the breeding season in the nesting area could be attributed to the following factors. 4.0pt

Factor	True	False
A. Guano deposition		
B. Soil pH		
C. Ambient temperature		
D. Soil disturbance due to nesting		

Q.2.4.b The effects of guano deposition on arthropod abundance can be evaluated only by analyzing Trap from all four experimental situations. 1.0pt

True	
False	

Q.2.5 Conduct a t-test to determine if there is a significant difference in ant abundance before and after the breeding of cormorants in the nesting area, using the following formula:

$$t = \frac{(x_1 - x_2)}{\sqrt{\frac{(SD_1)^2}{n_1} + \frac{(SD_2)^2}{n_2}}}$$

t	Test statistic
x_1 and x_2	The means of Samples 1 and 2 respectively
SD_1 and SD_2	Standard deviations of Samples 1 and 2 respectively
n_1 and n_2	The sample sizes of Samples 1 and 2 respectively

Q.2.5.a Put the t value, rounded to two decimal places 2.0pt

Q.2.5.b Given that the degrees of freedom for this test is 12, use Table 3 below to determine whether your t-test shows a statistically significant difference in ant abundance in the nesting area before and after cormorant breeding. 2.0pt

Is the t-value statistically significant at $p < 0.05$?

Yes	
No	

A	B					
	20% (0.20)	10% (0.10)	5% (0.05)	2% (0.02)	1% (0.01)	0.1% (0.001)
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.598
3	1.638	2.353	3.182	4.541	5.841	12.941
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.859
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.405
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.767
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.043	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.158	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

Table 3. Student's t table

A	Degree of freedom
B	Significance level

Task 3. Gular Fluttering in Socotra Cormorants

During the early part of the breeding season, cormorants experience temperatures as high as 40°C during the day. Birds that are heat-stressed carry out gular fluttering, continuously flexing a patch of bare skin on their neck.

Q.3.1 Observe gular fluttering in two individual cormorants, labelled 1 and 2, in Video 1.

Count the number of visible flutters in the gular of both Cormorants 1 and 2 in the suggested 10-second segments in Task 3-Video 1. Each gular flutter can be counted as one when it returns to its starting position. You can use the counter, which is provided to you and the timeline, available below the video, to aid in the counting process. You can also change the speed of the video, if you would like to.

Subsequently, calculate the gular fluttering rates of the two birds in terms of the number of flutters / minutes, in the table below. Calculate the mean and standard error of these rates for the two birds and round them to two decimal places.

Q.3.1.1 18.0pt

Cormorant 1				Cormorant 2		
Segment	Time observed (seconds)	Number of flutters	Flutters / minute	Time observed (seconds)	Number of flutters	Flutters / minute
1	0:35 – 0:45			0:11 – 0:21		
2	0:47 – 0:57			0:45 – 0:55		
3	1:04 – 1:14			0:56 – 1:06		

Number of flutters: 2 points each = 12 points
Flutters/minute: 1 point each = 6 points

Q.3.1.2 4.0pt

	Cormorant 1	Cormorant 2
	Flutters/minute	
Mean		
Standard error		

Mean: 1 point each = 2 points
Standard error: 1 point each = 2 points

Q.3.2 The average gular fluttering rate of a group of cormorants was calculated through the day and plotted in Figure 1 below. The air temperature surrounding the cormorants was also plotted.

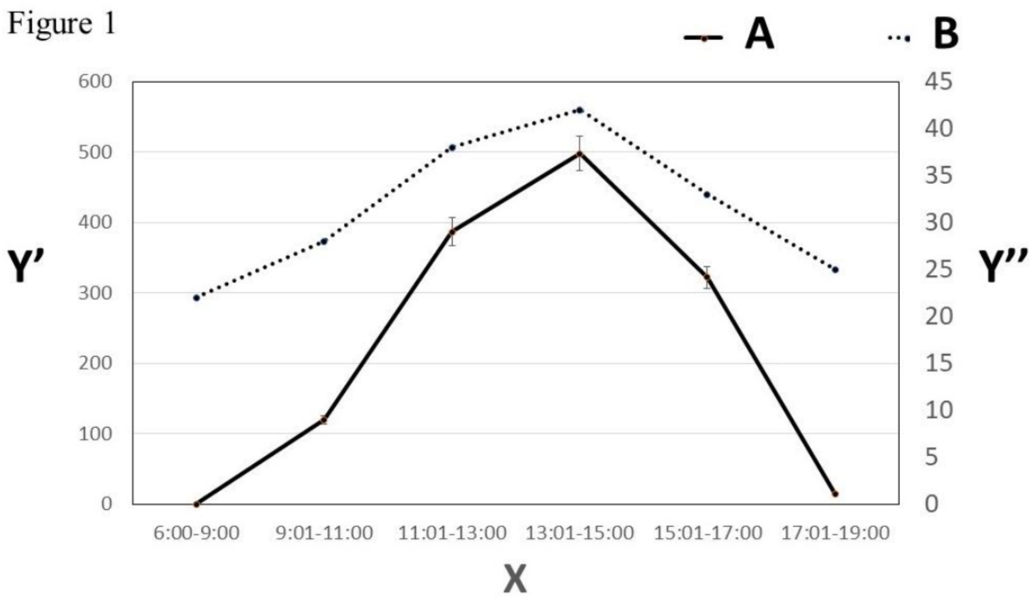


Figure 1

A	Gular fluttering rate
B	Temperature
X	Time of the day
Y'	Numbers per minute
Y''	Temperature ($^{\circ}C$)

True or False?

Q.3.2 5.0pt

Statement	**True**	**False**
A. Gular fluttering rate increases with the time of day		
B. Gular fluttering is likely to be an adaptive response to both extremely high and low temperatures		
C. Increasing temperature causes an increase in gular fluttering rate, as the latter enhances conductive heat loss from throat tissues		
D. Gular fluttering promotes water loss from throat tissues.		
E. Larger cormorant species spend a relatively greater proportion of time gular-fluttering for a particular temperature		

Q.3.3 In Video 1, that several cormorants were sitting in their nests while others were standing in them. The percentage of individual cormorants that were sitting or standing was recorded across the day (Figure 2 below).

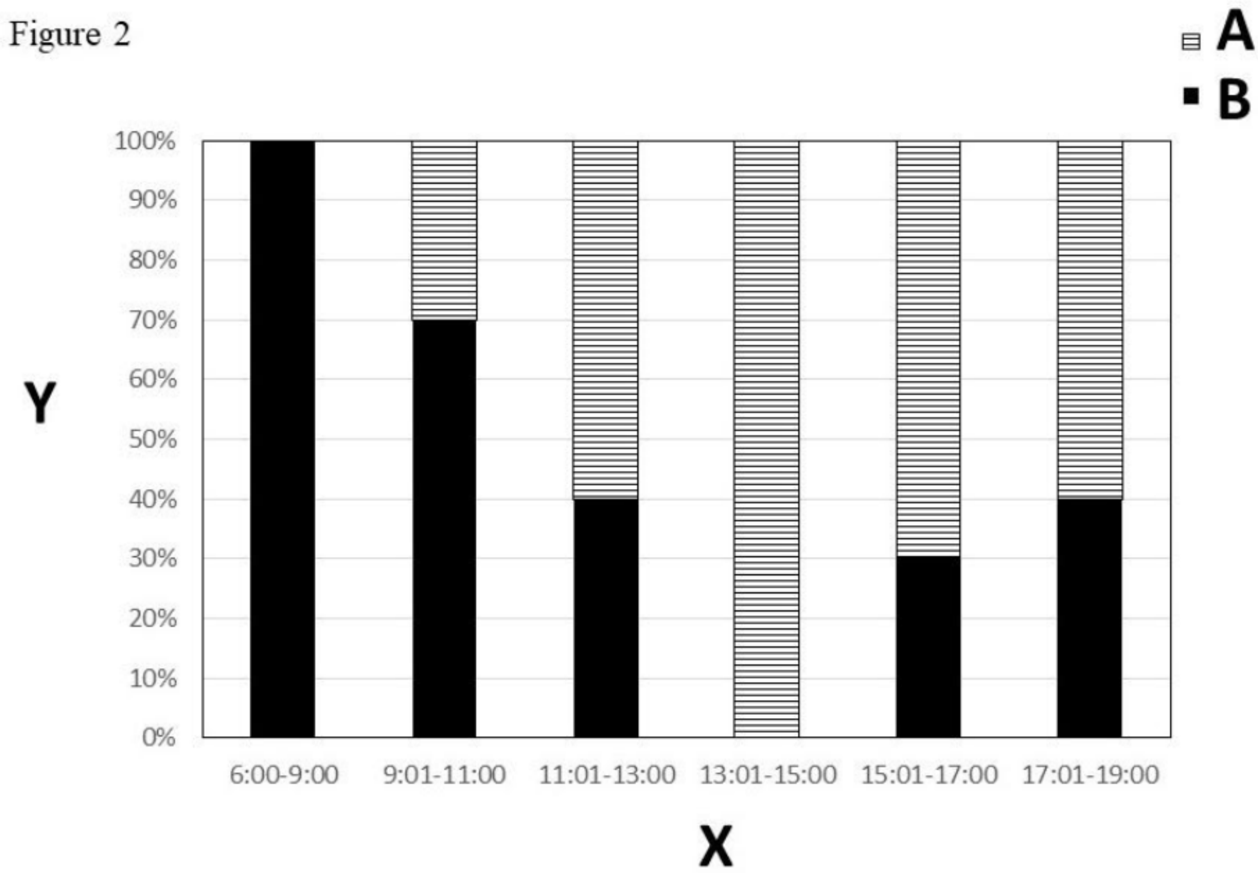


Figure 2

A	Standing
B	Sitting
X	Time of day
Y	Proportion of individuals

True or False?

Q.3.3 5.0pt

Statement	True	False
A. Cormorants prefer to sit in their nests, during morning hours to protect their chicks from predators, who are typically active at that time		
B. Gular fluttering rates are likely to be high in individual cormorants that spend more time sitting in their nests than do their standing neighbours		
C. Standing allows for better airflow between the legs of cormorants, promoting more efficient loss of heat from the body		
D. Cormorants stand in their nests to prevent heat loss from the bodies of their chicks		
E. Parent cormorants are often likely to trade-off thermoregulatory demands against offspring survival when they adopt certain body postures in their nests		

Task 4. Inter- and Intrasexual Communication in Fan-Throated Lizards

The superb fan-throated lizard *Sarada superba*, discovered in 2016, lives in semi-arid habitats on a single mountain plateau of the Western Ghats mountains, a global biodiversity hotspot, in India. The males have a tricoloured dewlap, shown below – an extendible flap of skin, usually folded under the throat – with blue, black and orange patches, which they flap during social interactions. The females have a rudimentary white dewlap.

The objective of this task will be to extract, analyze and interpret behavioral data from social interactions of individual lizards of this species.

You have been given four videos of male lizards in the wild. In each of two videos – Task 4-Videos 1 and 2 – a male is socially communicating with another male, who has been introduced into his territory in a transparent box. In each of the two other videos – Task 4-Videos 3 and 4 – a male is communicating with a female, although you cannot see the box with the female in the video frame.

None of the lizards were harmed during these experiments, and the boxed lizards were quickly returned to their own territories.

Q.4.1 Count the number of dewlap flaps made by the males in these four videos and calculate the number of flaps per minute in each case. Only count the displays of the two free-ranging males in Task 4-Videos 1 and 2; do **not** count that of the boxed males. In all videos, only count flaps that completely extend the dewlap. For examples of such complete extension, see Second 3 in Task 4-Video 2 or Second 8 in Task 4-Video 3. For partial extensions, which should **not** be counted, see Second 8 in Task 4-Video 1 or Seconds 12 and 13 in Task 4-Video 3.

Fill in the table below.

Q.4.1.1 14.0pt

	Number of flaps	Time observed (seconds)	Flaps/ minute
Social communication with males			
Video 1			
Video 2			
Social communication with females			
Video 3			
Video 4			

Number of flaps observed: 2 points each = 8 points
Time observed: 0.5 point each = 2 points
Flaps/minutes calculated: 1 point each = 4 points

Q.4.2 Use the chi-square test for goodness of fit to determine whether the difference in the mean rate of dewlap flaps per minute, displayed by male lizards to other males versus that to females is significantly different from the null hypothesis that there are no differences in such displays to the two sexes.

To carry out the test, fill in the appropriate values in the table below and compute the chi-square statistic by using the following formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- χ^2 is the chi-square statistic,
- O_i is the number of observations of type *i*,
- E_i is the expected count of type *i.*

Q.4.2.1 7.0pt

	Mean rate of dewlap flaps/minute, when communicating with:	
	Males	**Females**
Observed		
Expected		
Chi-square statistic, χ^2 rounded to two decimal places		
Degrees of freedom		

Observed: 1 point each = 2 points
Expected: 1 point each = 2 points
Chi-square: 2 points
Degree of freedom = 1 point

Use the table below to determine whether your chi-square test shows a statistically significant difference in the mean rate of dewlap flaps displayed by males to other males versus that to females.

Q.4.2.2 Is the chi-square value statistically significant at $p < 0.05$? 1.0pt

Yes	
No	

Critical values of the Chi-square distribution with d degrees of freedom

d	0.05	0.01	0.001	*d*	0.05	0.01	0.001
1	3.841	6.635	10.828	11	19.675	24.725	31.264
2	5.991	9.210	13.816	12	21.026	26.217	32.910
3	7.815	11.345	16.266	13	22.362	27.688	34.528
4	9.488	13.277	18.467	14	23.685	29.141	36.123
5	11.070	15.086	20.515	15	24.996	30.578	37.697
6	12.592	16.812	22.458	16	26.296	32.000	39.252
7	14.067	18.475	24.322	17	27.587	33.409	40.790
8	15.507	20.090	26.125	18	28.869	34.805	42.312
9	16.919	21.666	27.877	19	30.144	36.191	43.820
10	18.307	23.209	29.588	20	31.410	37.566	45.315

Q.4.3 True or False?

Q.4.3 2.0pt

Statement	**True**	**False**
A. Strong male–male competition could promote sexual selection in these lizards and the capability to repeatedly flap the dewlap could signal a male's relatively greater fighting ability to potential competitors		
B. Enhanced dewlap-flapping towards other males could be a mechanism to facilitate species recognition when closely related species are sympatric, that is, living in the same area		

Q.4.4 Superb fan-throated lizards exhibit sexual dimorphism, with the males possessing tricoloured dewlaps while those of the females are white.

True or False?

Q.4.4 2.0pt

Statement	**True**	**False**
A. The evolution of the tricoloured dewlap in the males could be attributed to the differential response of females and males to the differently coloured patches on the male dewlap		
B. The multiple colours of the dewlap could serve to improve signal detection for the receivers of the signal		

Q.4.5 Why do males flap their dewlap, rather than continuously keep it extended?

True or False?

Q.4.5 2.0pt

Statement	**True**	**False**
A. Leaving the dewlap extended is much more energetically demanding than flapping it		
B. Leaving it extended would attract more predators than flashing it		

Ecology and Ethology (103 Points)

MARK YOUR ANSWERS IN THIS ANSWER SHEET

ONLY THIS ANSWER SHEET WILL BE EVALUATED

Task 1. Mark-Release-Recapture Method to Estimate Population Size

A.1.1 (2.0 pt)

A.1.2 (3.0 pt)

Statement	True	False
A.		
B.		
C.		

A.1.3.1 (4.0 pt)

Marking event	Numbers of individuals captured with or without a mark (C)	Numbers of marked individuals recaptured (R)	Total number individuals newly marked	Total number of individuals captured and marked (M)	Estimated total population size (N)
1	20	-	20	20	-
2	20				
3	20				
4	20				
5	20				
Total	100	X	X	X	X

A.1.3.2 (1.0 pt)

Mean population estimate	

A.1.3.3 (1.0 pt)

Standard error of the estimate	

Q.1.3.4 (1.0 pt)

	The final estimate	The mean of the estimates
A.		

A.1.3.5 (5.0 pt)

Factor	True	False
A.		
B.		
C.		
D.		
E.		

Task 2. Response of Terrestrial Invertebrates to Guano Enrichment from Nesting Cormorants

A.2.1 (8.0 pt)

Nesting Area - Before Breeding Season				
	Beetles	Spiders	Ants	Ticks
Mean				
Standard deviation				

A.2.2.1 (2.0 pt)

A.2.2.2 (2.0 pt)

A.2.3.1 (1.0 pt)

True	
False	

A.2.3.2 (1.0 pt)

True	
False	

A.2.3.3 (1.0 pt)

True	
False	

A.2.3.4 (1.0 pt)

True	
False	

A.2.3.5 (1.0 pt)

True	
False	

A.2.4.a (4.0 pt)

Factor	True	False
A.		
B.		
C.		
D.		

A.2.4.b (1.0 pt)

True	
False	

A.2.5.a (2.0 pt)

A.2.5.b (2.0 pt)

Yes	
No	

Task 3. Gular Fluttering in Socotra Cormorants

A.3.1.1 (18.0 pt)

Cormorant 1				Cormorant 2		
Segment	Time observed (seconds)	Number of flutters	Flutters / minute	Time observed (seconds)	Number of flutters	Flutters / minute
1	0:35 – 0:45			0:11 – 0:21		
2	0:47 – 0:57			0:45 – 0:55		
3	1:04 – 1:14			0:56 – 1:06		

A.3.1.2 (4.0 pt)

	Cormorant 1	Cormorant 2
	Flutters/minute	
Mean		
Standard error		

A.3.2 (5.0 pt)

Statement	True	False
A.		
B.		
C.		
D.		
E.		

A.3.3 (5.0 pt)

Statement	True	False
A.		
B.		
C.		
D.		
E.		

Task 4. Inter-and Intrasexual Communication in Fan-Throated Lizards

A.4.1.1 (14.0 pt)

	Number of flaps	Time observed (seconds)	Flaps / minute
Social communication with males			
Video 1			
Video 2			
Social communication with females			
Video 3			
Video 4			

A.4.2.1 (7.0 pt)

	Mean rate of dewlap flaps / minute, when communicating with:	
	Males	**Females**
Observed		
Expected		
Chi-square statistic, χ^2 , rounded to two decimal places.		
Degrees of freedom		

A.4.2.2 (1.0 pt)

Yes	
No	

A.4.3 (2.0 pt)

Statement	True	False
A.		
B.		

A.4.4 (2.0 pt)

Statement	True	False
A.		
B.		

A.4.5 (2.0 pt)

Statement	True	False
A.		
B.		

Experiment 3: Ecology and Ethology **103 points**

IBO 2023, UAE

Sabir Bin Muzaffar and Anindya Sinha

Task 1. Mark-Release-Recapture Method to Estimate Population Size **17 points**

Mark-release-recapture methods are used to estimate the population size of mobile animals. A known number of individuals of a population are captured and marked, with a ring or an ear tag, which has been tested to not cause any mortality, and then released back into the population. When individuals of the same population are captured again, after giving the individuals sufficient time to mix with the rest of the population, the researchers expect to recapture a certain proportion of the previously marked individuals. The marked individuals represent a proportion of the total population that can be used to estimate the total population of individuals.

Q1.1 Given

- "N" is the estimated total population size,
- "M" is the total number of individuals captured and marked,
- "C" is the number of individuals captured with or without a mark,
- "R" is the number of marked individuals recaptured.

Write down the formula by which "N" can be estimated.

Answer: $N = (M * C) / R$

2 points

Explanation:

The ratio of the number of marked individuals that are recaptured (or captured the next time) to the total number of individuals marked should be equal to the ratio of the total number of individuals captured the next time to the estimated total population size,

or, $R / M = C / N$

or, $N = (M * C) / R$

Q1.2 If no marked individuals are recaptured in the above process, we conclude that:

Statement	True	False
The population is infinitely large		√
The marked individuals have all died		√
It is necessary to mark more individuals and make fresh attempts to estimate the size of the population	√	

3 points

Explanation:

Q1.2.1 The population could be large, if not infinitely large, but it can definitely be estimated with effort.

Q1.2.2 It is very unlikely that all the marked individuals have died.

Q1.2.3 This statement is obviously true in principle.

Q1.3 In an experiment, you are given a bag, containing a known number of beads, to simulate a population of animals.

(1) Take out 20 beads randomly, mark them with the marker provided and then return the marked beads to the bag.

(2) Shake the bag well to ensure that the beads are mixed properly.

(3) Take out another 20 beads randomly and note how many of them are already marked from your first capture.

(4) Mark the unmarked beads from this second capture and then return all the newly marked and previously marked beads back to the bag.

(5) Conduct this process for a total of five times. In each event, mark the **unmarked** beads that you capture and return all beads, so that the total number of marked beads increases with each repeat round.

(6) Fill the table below. Estimate the total population size, mean population estimate and the standard error of the estimate to one decimal place.

Standard deviation, $$\mathrm{SD} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Standard error, $\mathrm{SE} = \frac{\mathrm{SD}}{\sqrt{n}}$

- x_i is the value of the i^{th} point in the data set,
- $\bar{x}$ is the mean of the data set,
- n is the number of data points in the data.

Q.1.3.1

Marking event	Number of individuals captured with or without a mark (C)	Number of marked individuals recaptured (R)	Number of individuals newly marked	Total number of individuals captured and marked (M)	Estimated total population size (N)
1	20	–	20	20	–
2	20	3	17	37	133.3
3	20	8	12	49	92.5
4	20	4	16	65	245.0
5	20	6	14	79	216.7
Total	100				
Mean population estimate					171.9
Standard error of the estimate					35.5

Each correct row, Events 2 to 5 = **4 points**

(Each correct cell, with the correct formula for N, from Events 2 to 5, gets 0.25 point = **4 points**

Each correct cell, with an incorrect formula for N, from Events 2 to 5, gets 0.25 point = **3 points**)

Q.1.3.2 Calculated mean population estimate = **1 point**

Q.1.3.3 Calculated standard error of the estimate = **1 point**

The values in Column 3 are representative, obtained during a trial test while the other values have been calculated.

Q1.3.4 Mark with a cross (X) the correct answer:

	The final estimate	The mean of the estimates
Which would finally be a better estimate of the total population size?	√	

1 point

Explanation:

The final estimate will always be a better estimate of the total population size, as reliability increases statistically with an increasing number of individuals becoming marked in the population. The mean estimate, however, would increase the statistically unreliable earlier estimates and hence, more likely to be away from the actual population size.

In a real population of marked animals, such as the Arabian red fox *Vulpes vulpes arabica*, shown above, various factors can introduce errors in the estimated population size.

Q.1.3.5Considering the process described above, indicate which of the following factors would introduce an error in the population estimate?

Factor	True	False
A. The same individual fox is trapped more frequently than others	√	
B. The collar on an individual fox drops off	√	
C. The collared foxes do **not** emigrate during the experiments		√

D. A very small proportion of the fox population is marked at the end of all mark-recapture events	√	
E. The recaptures are carried out in very rapid succession	√	

5 points

Explanation:

A. Certain individuals are trap-happy and are attracted repeatedly to the trap. This would obviously be a deviation from the principle that each capture of an individual is random and independent of the previous captures. The estimated population size would thus decrease from its actual value.

B. A collar dropping off from an individual would lead to the impression that this individual, if trapped again, was previously uncaptured. The estimated population size would thus increase from its actual value.

C. An important assumption of this method of population size estimation is that the population is 'closed' during the exercise and that there are no deaths, immigration or emigration of individuals during the relatively short period of time over which the estimation is carried out.

D. A very small proportion of individuals being marked would lower the probability of such marked individuals being recaptured and hence, lead to the false conclusion that the population is larger than it actually is.

E. It is expected that the recaptures should be carried out in quick succession as this would then not violate the assumption of the population being 'closed' during the estimation exercise.

Task 2. Response of Terrestrial Invertebrates to Guano Enrichment from Nesting Cormorants

26 points

The Socotra cormorant *Phalacrocorax nigrogularis* is a threatened aquatic bird, endemic to the Arabian Gulf and found off the southeast coast of the Arabian peninsula. They usually nest in colonies of tens-to-thousands of nesting pairs. In the breeding season, between August and December, they build cup-shaped nests on the ground, lay eggs, incubate them, and eventually tend to their hatchlings, including protecting them from predators, such as the nocturnal Arabian red fox.

Large colonies of breeding Socotra cormorants deposit thousands of tons of faeces (guano), which alter soil chemistry, greatly increasing nutrients, such as nitrogen. This has variable impacts on the soil flora and fauna. Populations of some species increase due to the guano enrichment, others decrease and some remain unaffected.

You are given data showing invertebrates captured in pitfall traps that were placed in a large cormorant colony. Pitfall traps are small, cup-shaped containers, buried up to their rim in the ground, with a liquid preservative in them to prevent insects from escaping when they fall in.

The image above shows a cormorant nesting area (left) and a non-nesting area (right). Pitfall traps (blue dots) were placed in both areas for a period of one week one month before the nesting period. The pitfall traps were put back in the same places for another week one month after the nesting period. The trap contents were collected at the end of the one-week periods. All the collected invertebrates were then identified and counted. These taxa included Coleoptera (beetles), Araneae (spiders), Formicidae (ants) and Argasidae (soft ticks). Their numbers have been reported in Tables 1 and 2 below.

Table 1

Nesting Area – Before Breeding Season					Nesting Area – After Breeding Season				
Pitfall Traps	Beetles	Spiders	Ants	Ticks	Pitfall Traps	Beetles	Spiders	Ants	Ticks
Trap 1	33	13	10	2	Trap 1	4	1	1	38
Trap 2	27	17	11	5	Trap 2	7	2	1	41
Trap 3	29	18	9	1	Trap 3	4	2	2	32
Trap 4	19	18	12	3	Trap 4	3	3	1	45
Trap 5	22	14	9	4	Trap 5	1	1	1	28
Trap 6	26	19	8	4	Trap 6	8	4	3	27
Trap 7	31	15	11	3	Trap 7	5	6	1	31
Trap 8	26	14	13	2	Trap 8	6	2	1	33
Trap 9	21	19	12	7	Trap 9	2	3	1	37
Trap 10	28	20	11	2	Trap 10	1	3	2	41
Mean	26.2	16.7	10.6	3.3	Mean	4.1	2.7	1.4	35.3
Standard deviation	4.4	2.5	1.6	1.8	Standard deviation	2.4	1.5	0.7	6.0

Table 2

Non-Nesting Area – Before Breeding Season					Non-Nesting Area – After Breeding Season				
Pitfall Traps	Beetles	Spiders	Ants	Ticks	Pitfall Traps	Beetles	Spiders	Ants	Ticks
Trap 1	25	15	7	3	Trap 1	3	13	14	1
Trap 2	22	16	10	2	Trap 2	2	17	11	4
Trap 3	20	18	11	1	Trap 3	1	19	12	2
Trap 4	18	18	13	2	Trap 4	1	13	10	3
Trap 5	18	19	10	6	Trap 5	5	14	10	3
Trap 6	17	21	11	1	Trap 6	4	13	12	5
Trap 7	19	14	13	1	Trap 7	2	18	11	4
Trap 8	15	12	9	2	Trap 8	2	16	12	4
Trap 9	20	13	13	5	Trap 9	7	15	10	1
Trap 10	21	13	12	1	Trap 10	1	13	9	2
Mean	19.5	15.9	10.9	2.4	Mean	2.8	15.1	11.1	2.9
Standard deviation	2.8	3.0	2.0	1.8	Standard deviation	2.0	2.3	1.4	1.4

Q2.1 Calculate the mean and standard deviation of the abundance of beetles, spiders, ants and ticks in the Nesting Area – Before Breeding Season and write them, rounded to one decimal place, in Table 1 above.

Each cell has 1 point each = **8 points**

Q2.2 Calculate the Shannon-Wiener Diversity Index (H') for Trap 1 of pitfall traps in the Nesting Area – Before Breeding Season and in the Nesting Area – After Breeding Season, by using the following formula:

$$H' = -\sum_{i=1}^{S} p_i \ln p_i$$

where P_i, the proportion of taxon '*i*' relative to the total number of taxa, is computed, and then multiplied by $\ln(P_i)$, the natural logarithm of this proportion.

Give the answer, rounded to two decimal places, in the boxes below:

Q.2.2.1

Nesting Area – Before Breeding Season =

Correct answer is 1.08

Q.2.2.2

Nesting Area – After Breeding Season =

Correct answer is 0.52

2 + 2 points = **4 points**

Q2.3 True or false?

Q.2.3.1 The diversity of taxa, as measured by H', is higher in the Nesting Area – After Breeding Season.

☐ ☐

True False √

Q.2.3.2 In general, the H' for a habitat, with only a single species present, is 0.

True ☐ √ False ☐

Q.2.3.3 H' depends on both species richness, which is the number of species in a particular habitat, and on species evenness, which is the relative abundance of each of the species in that habitat.

True ☐ √ False ☐

Q.2.3.4 The H' of a particular habitat decreases as species evenness, in that habitat, increases.

True ☐ False ☐ √

Q.2.3.5 A habitat with a relatively higher H' value, compared to another habitat, is likely to have lower interspecies competition for resources.

True ☐ False ☐ √

5 points

Explanations:

Q2.3.1 This statement can be evaluated by the calculation of the Index.

Q2.3.2 If there is only a single species in a community, its proportion in the community is 1 and ln of 1 is 0. And hence, the value of the Index becomes 0.

Q2.3.3 It is quite evident from the formulation of the Index that it considers both, the number of species living in a habitat or its species richness and the relative abundance of these species, that is, its species evenness.

Q2.3.4 It is also quite clear from the formulation of the Index that its value increases as the species evenness of the habitat increases.

Q2.3.5 It should be noted that relatively higher values of the Index indicate greater species diversity in the habitat while lower values indicate a lower diversity. Hence, it is expected that interspecies competition for resources will be higher when species diversity is higher in a habitat.

Q2.4 True or false?

Q.2.4.aThe observed differences in arthropod abundance before and after the breeding season in the nesting area could be attributed to the following factors.

Factor	**True**	**False**
A. Guano deposition	√	
B. Soil pH	√	
C. Ambient temperature	√	
D. Soil disturbance due to nesting	√	

4 points

Q.2.4.b The effects of guano deposition on arthropod abundance can be evaluated only by analysing traps from all four experimental situations.

True ☑ False ☐

1 point

Explanations:

Q2.4 It is possible that differences in arthropod abundance before and after the breeding season could be sensitive to all the factors mentioned while it is true that the effects of guano deposition on arthropod abundance can be evaluated only by analysing samples from all four experimental situations.

Q2.5 Conduct a t-test to determine if there is a significant difference in ant abundance before and after the breeding of cormorants in the nesting area, using the following formula:

$$t = \frac{(x_1 - x_2)}{\sqrt{\frac{(SD_1)^2}{n_1} + \frac{(SD_2)^2}{n_2}}}$$

- t is the test statistic,
- x_1 and x_2 are the means of Samples 1 and 2 respectively
- SD_1 and SD_2 are the standard deviations of Samples 1 and 2 respectively
- n_1 and n_2 are the sample sizes of Samples 1 and 2 respectively.

Q.2.5.a Put the t value, rounded to two decimal places:

☐

Correct answer = 16.66 – 16.86

2 points

Q.2.5.b Given that the degrees of freedom in this test is 12, use Table 3 below to determine whether your t-test shows a statistically significant difference in ant abundance in the nesting area before and after cormorant breeding.

Is the t value statistically significant at $p < 0.05$?

Yes ☑ No ☐

2 points

Degrees of freedom	Significance level					
	20% (0.20)	10% (0.10)	5% (0.05)	2% (0.02)	1% (0.01)	0.1% (0.001)
1	3.078	6.314	12.706	31.821	63.657	636.619
2	1.886	2.920	4.303	6.965	9.925	31.598
3	1.638	2.353	3.182	4.541	5.841	12.941
4	1.533	2.132	2.776	3.747	4.604	8.610
5	1.476	2.015	2.571	3.365	4.032	6.859
6	1.440	1.943	2.447	3.143	3.707	5.959
7	1.415	1.895	2.365	2.998	3.499	5.405
8	1.397	1.860	2.306	2.896	3.355	5.041
9	1.383	1.833	2.262	2.821	3.250	4.781
10	1.372	1.812	2.228	2.764	3.169	4.587
11	1.363	1.796	2.201	2.718	3.106	4.437
12	1.356	1.782	2.179	2.681	3.055	4.318
13	1.350	1.771	2.160	2.650	3.012	4.221
14	1.345	1.761	2.145	2.624	2.977	4.140
15	1.341	1.753	2.131	2.602	2.947	4.073
16	1.337	1.746	2.120	2.583	2.921	4.015
17	1.333	1.740	2.110	2.567	2.898	3.965
18	1.330	1.734	2.101	2.552	2.878	3.922
19	1.328	1.729	2.093	2.539	2.861	3.883
20	1.325	1.725	2.086	2.528	2.845	3.850
21	1.323	1.721	2.080	2.518	2.831	3.819
22	1.321	1.717	2.074	2.508	2.819	3.792
23	1.319	1.714	2.069	2.500	2.807	3.767
24	1.318	1.711	2.064	2.492	2.797	3.745
25	1.316	1.708	2.060	2.485	2.787	3.725
26	1.315	1.706	2.056	2.479	2.779	3.707
27	1.314	1.703	2.052	2.473	2.771	3.690
28	1.313	1.701	2.048	2.467	2.763	3.674
29	1.311	1.699	2.043	2.462	2.756	3.659
30	1.310	1.697	2.042	2.457	2.750	3.646
40	1.303	1.684	2.021	2.423	2.704	3.551
60	1.296	1.671	2.000	2.390	2.660	3.460
120	1.289	1.658	1.980	2.158	2.617	3.373
∞	1.282	1.645	1.960	2.326	2.576	3.291

Table 3. Student's t table

Task 3. Gular Fluttering in Socotra Cormorants **32 points**

During the early part of the breeding season, cormorants experience temperatures as high as 40ºC during the day. Birds that are heat-stressed carry out gular fluttering, continuously flexing a patch of bare skin on their neck back and forth.

Q3.1 Observe gular fluttering in two individual cormorants, labelled 1 and 2, in Video 1.

The image above is a screenshot from the Second 39 in Video 1.

Count the number of visible flutters in the gular of both Cormorants 1 and 2 in the suggested 10-second segments in Task 3–Video 1. Each gular flutter can be counted as one when it returns to its starting position. You can use the counter, which is provided to you and the timeline, available below the video, to aid in the counting process. You can also change the speed of the video, if you would like to.

Subsequently, calculate the gular fluttering rates of the two birds in terms of the number of flutters / minute in the table below. Calculate the mean and standard deviation of these rates for the two birds and round them to two decimal places.

Q.3.3.1

	Cormorant 1			Cormorant 2		
Segment	Time observed (seconds)	Number of flutters	Flutters / minute	Time observed (seconds)	Number of flutters	Flutters / minute
1	0:35 – 0:45	53-63	318-378	0:11 – 0.21	42-52	262-312
2	0:47 – 0:57	61-71	366-426	0:45 – 0:55	35-45	210-270
3	1:04 – 1:14	57-67	342-402	0:56 – 1:06	51-61	306-366
Mean			332-402			259-316
Standard error			13.86			27.78

18 points

Number of flutters: 2 points each = 12 points

Flutters/minute: 1 point each = 6 points

Q.3.1.2

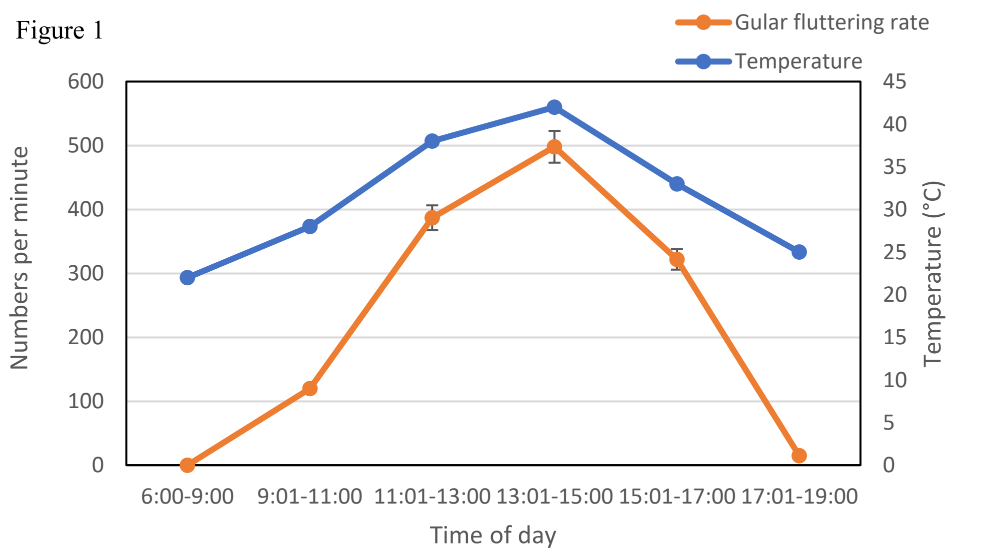
Mean: 1 point each = 2 points

Standard error: 1 point each = 2 points

The values of the observed number of flutters – Columns 3 and 6 – are representative and were obtained during trial tests. The rest of the values have been calculated.

The average gular fluttering rate of a group of cormorants was calculated through the day and plotted in Figure 1 below. The air temperature surrounding the cormorants was also plotted.

Figure 1



True or false?

Q.3.2

Statement	True	False
A. Gular fluttering rate increases with the time of day		√
B. Gular fluttering is likely to be an adaptive response to both extremely high and low temperatures		√
C. Increasing temperature causes an increase in gular fluttering rate, as the latter enhances conductive heat loss from throat tissues		√
D. Gular fluttering promotes water loss from throat tissues	√	
E. Larger cormorant species spend a relatively greater proportion of time gular-fluttering for a particular temperature	√	

5 points

Explanations:

A. Gular fluttering initially increases with time of day and then decrease after 13:00h.
B. Gular fluttering reduces significantly at low temperatures.
C. Increasing temperature does indeed increase gular fluttering rates but this is because such fluttering enhances convective – not conductive – heat loss from throat tissues.
D. Gular fluttering promotes water loss from throat tissues.
E. Larger-bodied cormorant species have relatively lower surface area: volume ratio and thus, in order to increase rates of heat loss, need to spend relatively greater proportions of time gular-fluttering at a particular temperature; they typically also have a lower temperature threshold to initiate gular fluttering.

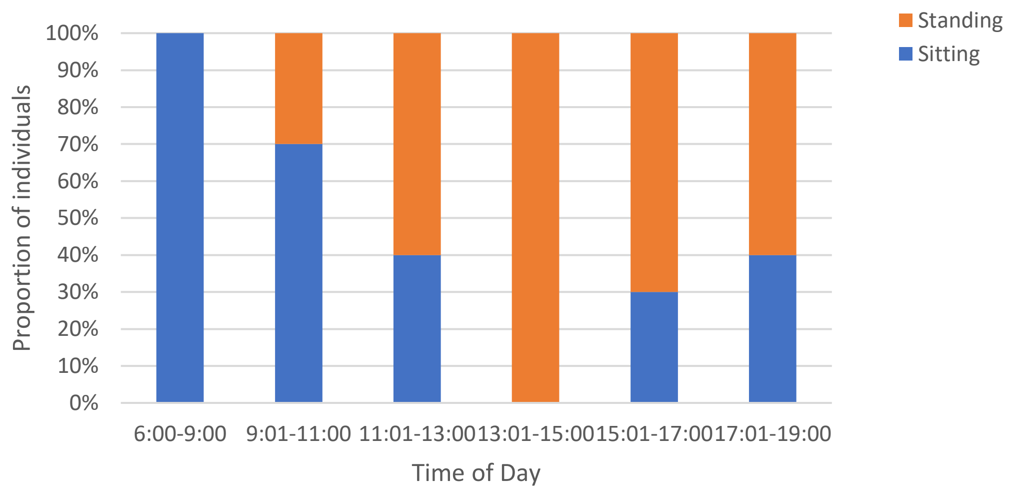
References:

Campbell 2014 – *Effects of Temperature on Gular Fluttering and Evaporative Water Loss in Four Sympatric Cormorants in Southern Africa*. Master's Thesis, Percy FitzPatrick Institute of African Ornithology, University of Cape Town, Cape Town, South Africa.

Cook et al. 2020 – Parenting in a Warming World - Thermoregulatory Responses to Heat Stress in an Endangered Seabird. *Conservation Physiology* 8(1): coz109.

Q3.3 In Video 1, several cormorants were sitting in their nests while others were standing in them. The percentage of individual cormorants that were sitting or standing was recorded across the day (Figure 2 below).

Figure 2



True or false?

Statement	**True**	**False**
A. Cormorants prefer to sit in their nests during morning hours to protect their chicks from predators, who are typically active at that time		√
B. Gular fluttering rates are likely to be high in individual cormorants that spend more time sitting in their nests than do their standing neighbours	√	
C. Standing allows for better airflow between the legs of cormorants, promoting more efficient loss of heat from the body	√	
D. Cormorants stand in their nests to prevent heat loss from the bodies of their chicks		√

E. Parent cormorants are often likely to trade-off thermoregulatory demands against offspring survival when they adopt certain body postures in their nests	√	

5 points

Explanations:

A. Cormorants prefer to sit, rather than stand, in their nests during morning hours, as the need to lose heat is not so high at that time but not necessarily to protect their chicks from predators, who are typically **not** active at that time.

B. Gular fluttering rates are likely to be high in cormorants that spend more time sitting in their nests, as this posture relatively reduces their surface area for heat loss, in comparison to standing cormorants.

C. Standing does indeed allow for better airflow between the legs of cormorants, promoting more efficient loss of heat from the body.

D. Cormorants stand in their nests to increase heat loss from their own bodies; if they had to reduce heat loss from the bodies of their chicks, they would need to cover them and protect them from the atmosphere.

E. Parent cormorants are indeed likely to trade-off thermoregulatory demands against offspring survival when they adopt certain body postures in their nests. These could include, for example, standing up as this would enhance heat loss from their bodies at relatively higher temperatures but, at the same time, expose their chicks to increased predation pressures.

References:

Campbell G. 2014. *Effects of Temperature on Gular Fluttering and Evaporative Water Loss in Four Sympatric Cormorants in Southern Africa*. Master's Thesis, Percy FitzPatrick Institute of African Ornithology, University of Cape Town, Cape Town, South Africa.

Cook TR et al. 2020. Parenting in a warming world: Thermoregulatory responses to heat stress in an endangered seabird. *Conservation Physiology* 8(1): coz109.

Task 4. Inter- and Intrasexual Communication in Fan-Throated Lizards 28 points

The superb fan-throated lizard *Sarada superba*, discovered in 2016, lives in semi-arid habitats on a single mountain plateau of the Western Ghats mountains, a global biodiversity hotspot, in India. The males have a tricoloured dewlap, shown below – an extendible flap of skin, usually folded under the throat – with blue, black and orange patches, which they flap during social interactions. The females have a rudimentary white dewlap.

The objective of this task will be to extract, analyse and interpret behavioural data from social interactions of individual lizards of this species.

You have been given four videos of male lizards in the wild. In each of two videos – Task 4–Videos 1 and 2 – a male is socially communicating with another male, who has been introduced into his territory in a transparent box. In each of the two other videos – Task 4–Videos 3 and 4 – a male is communicating with a female, although you cannot see the box with the female in the video frame.

None of the lizards were harmed during these experiments, and the boxed lizards were quickly returned to their own territories.

Q4.1 Count the number of dewlap flaps made by the males in these four videos and calculate the number of flaps per minute in each case. Only count the displays of the two free-ranging males in Task 4–Videos 1 and 2; do **not** count that of the boxed males. In all videos, only count flaps that completely extend the dewlap. For examples of such complete extension, see Second 3 in Task 4–Video 2 or Second 8 in Task 4–Video 3. For partial extensions, which should **not** be counted, see Second 8 in Task 4–Video 1 or Seconds 12 and 13 in Task 4–Video 3.

Fill in the table below.

Q.4.1.1

	Number of flaps	Time observed (seconds)	Flaps / minute
Social communication with males			
Video 1	6-10	20	18-30
Video 2	9-12	19	28-38
Social communication with females			
Video 3	10-15	59	10-15
Video 4	19-25	60	19-25

14 points

Number of flaps observed: 2 points each = 8 points

Time observed: 0.5 point each = 2 points

Flaps/minute calculated: 1 point each = 4 points

The values of the observed number of flaps – Column 2 – represent the range of values expected from observation, that of the time observed – Column 3 – are arbitrary while those of the flaps / minute – Column 4 – have been calculated.

Q4.2 Use the chi-square test for goodness of fit to determine whether the difference in the mean rate of dewlap flaps per minute, displayed by male lizards to other males versus that to females is significantly different from the null hypothesis that there are no differences in such displays to the two sexes.

To carry out the test, fill in the appropriate values in the table below and compute the chi-square statistic by using the following formula:

$$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$$

- χ^2 is the chi-square statistic,
- O_i is the number of observations of type i,
- E_i is the expected count of type i.

Q.4.2.1

	Mean rate of dewlap flaps / minute, when communicating with:	
	Males	Females
Observed	23-34	14-20
Expected	(Value for Males + Value for Females) / 2	(Value for Males + Value for Females) / 2
Chi-square statistic, χ^2, rounded to two decimal places	Calculated from the values above	
Degrees of freedom	1	

7 points

Observed: 1 point each = 2 points

Expected: 1 point each = 2 points

Chi-square = 2 points

Degrees of freedom = 1 point

Use the table below to determine whether your chi-square test shows a statistically significant difference in the mean rate of dewlap flaps displayed by males to other males versus that to females.

Q.4.2.2

Is the chi-square value statistically significant at $p < 0.05$?

☐ ☐

The answer could be true or false, depending on the observations

1 point

Critical values of the Chi-square distribution with *d* degrees of freedom

	Probability of exceeding the critical value						
d	0.05	0.01	0.001	*d*	0.05	0.01	0.001
1	3.841	6.635	10.828	11	19.675	24.725	31.264
2	5.991	9.210	13.816	12	21.026	26.217	32.910
3	7.815	11.345	16.266	13	22.362	27.688	34.528
4	9.488	13.277	18.467	14	23.685	29.141	36.123
5	11.070	15.086	20.515	15	24.996	30.578	37.697
6	12.592	16.812	22.458	16	26.296	32.000	39.252
7	14.067	18.475	24.322	17	27.587	33.409	40.790
8	15.507	20.090	26.125	18	28.869	34.805	42.312
9	16.919	21.666	27.877	19	30.144	36.191	43.820
10	18.307	23.209	29.588	20	31.410	37.566	45.315

Q4.3 True or false?

Statement	**True**	**False**
A. Strong male–male competition could promote sexual selection in these lizards and the capability to repeatedly flap the dewlap could signal a male's relatively greater fighting ability to potential competitors	√	
B. Enhanced dewlap-flapping towards other males could also be a mechanism to facilitate species recognition when closely related species are sympatric, that is, living in the same area		√

2 points

Explanations:

A. It is now well established in the literature on agamid lizards that body colouration and display behaviours have largely been promoted by intrasexual sexual selection, that is, intense competition between males for access to females.

B. Species recognition mechanisms typically evolve in sympatric species to ensure mate recognition and prevent non-productive cross-species reproduction.

Q4.4 Superb fan-throated lizards exhibit sexual dimorphism, with the males possessing tricoloured dewlaps while those of the females are white.

Statement	True	False
A. The evolution of the tricoloured dewlap in the males could be attributed to the differential response of females and males to the differently coloured patches on the male dewlap	√	
B. The multiple colours of the dewlap could serve to improve signal detection for the receivers of the signal	√	

2 points

Explanations:

A. It has been shown both observationally and empirically that the different colours on the male dewlap seem to have evolved by sexual selection, through the differential responses of the two sexes to these colours. Females preferentially respond to orange colour on male dewlaps in this species while males respond to blue and black, indicating that different colours may have evolved to target different receivers (Zambre and Thaker 2017).

B. It is now well established in the ethological literature across species, including insects, birds and mammals, that increasingly bright and contrasting colours improve their detection by the receivers of the signal.

Q4.5 Why do males flap their dewlap, rather than continuously keep it extended?

Statement	True	False
A. Leaving the dewlap extended is much more energetically demanding than flapping it		√
B. Leaving it extended would attract more predators than flashing it		√

2 points

A. Keeping a limb or appendage extended for a certain period of time is generally less energetically demanding than is repeatedly flapping it over the same period of time. This has been well established, for example, in bird flight, where gliding flight is less energetically expensive than in flapping flight.

B. It is now well known in insects, fish, reptiles, birds or mammals that sudden flaps or flashes of a brightly coloured appendage is more likely to attract an observer – both conspecific or heterospecific – than keeping it continuously exposed. This is also known as the startle effect. In this particular case, the lizards can be seen to be alert when flashing their dewlaps, possibly to also keep a lookout for potential predators.

References on agamid lizards:

Kamath A. 2016.Variation in display behavior, ornament morphology, sexual size dimorphism, and habitat structure in the fan-throated lizard (*Sitana*, Agamidae). *Journal of Herpetology* 50(3): 394-403

Zambre AM and Thaker M. 2017. Flamboyant sexual signals: Multiple messages for multiple receivers. *Animal Behaviour* 127: 197-203.